

Background & Motivation

- **Foundation Models in Medicine:** Vision-Language Models have revolutionized AI’s text and image processing, showing great promise for medical applications like automated radiology reporting, clinical decision support, and interactive QA.
- **The Data Dependency Challenge:** Driven by data quality and language coverage, medical AI thrives in high-resource languages with advanced datasets but stalls in low-resource languages due to severe data bottlenecks.
- **The Critical Gap for Bangla:** Though spoken by 278 million people globally, Bangla lacks credible medical AI benchmarks. Its sole public dataset lacks image documentation, expert validation, and clinical oversight, stalling local diagnostic AI development.
- **Motivation:** To enable equitable, reliable medical AI, we must build and evaluate models in linguistically diverse and clinically rigorous settings.

Research Question

How effective are current Large Vision-Language Models at performing Medical Visual Question Answering in Bangla, a low-resource language?

Contributions

- **First Bangla Medical VQA Dataset:** We introduce BanglaMedVQA, the first clinically validated Medical VQA benchmark for the Bangla language.
- **Expert-Verified Annotations:** The dataset comprises 7,000 diverse medical image-question-answer pairs, rigorously validated by qualified medical experts.
- **Comprehensive Model Evaluation:** We benchmark 9 state-of-the-art LVLMs, including Gemini, GPT-4, Claude and LLaMA.
- **Low-Resource Medical AI Baseline:** Models are evaluated on both Bangla and English datasets under Zero-Shot, CoT and LoRA fine tuning settings, establishing a strong baseline for low-resource medical AI research.

BANGLAMEDVQA Dataset

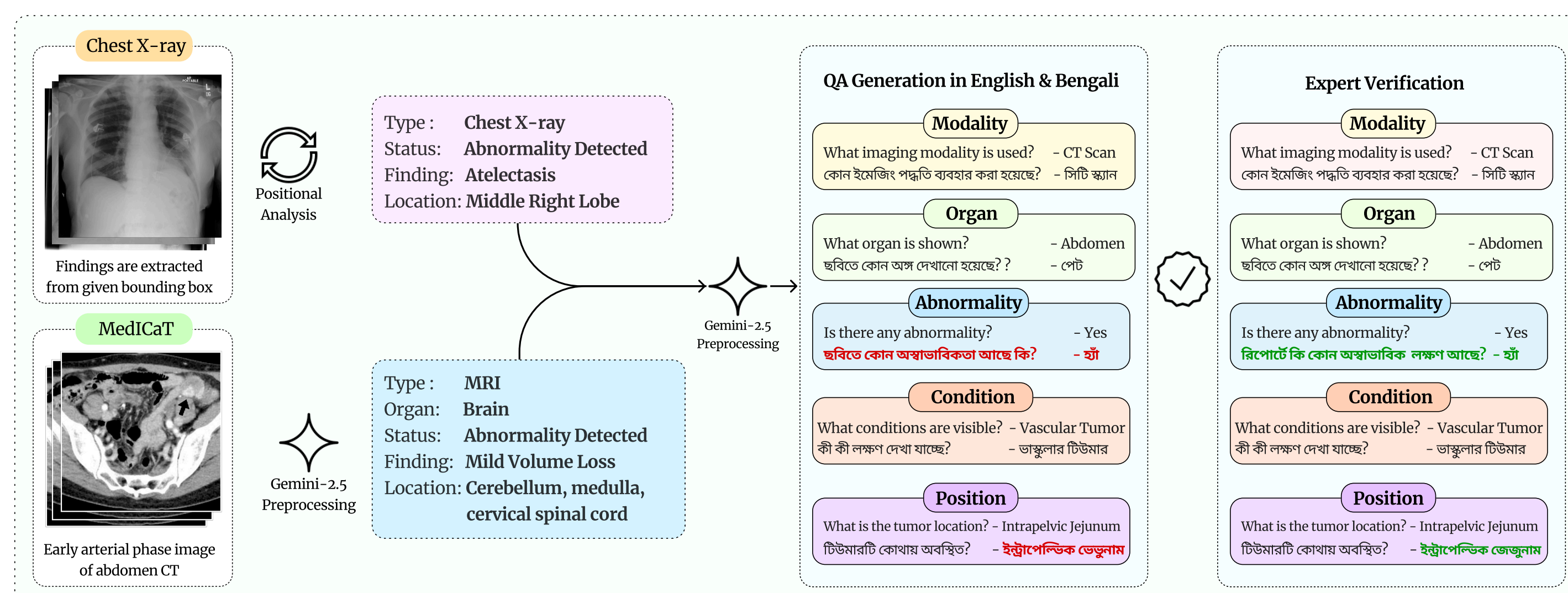


Figure 1. BanglaMedVQA dataset creation pipeline.

1. **Data Sources** To ensure clinical relevance and scale, we leveraged two validated biomedical datasets:

- **MedlCaT:** Provides detailed captions and finding descriptions.
- **ChestX-ray14:** Offers structured condition metadata and bounding box annotations.

2. **Generation Pipeline** We employed a two-stage LLM-driven pipeline to ensure linguistic and clinical consistency:

1. **Metadata Extraction:** Used **Gemini** (for captions) and a positional module (for bounding boxes) to extract standardized findings.
2. **QA Generation:** Generation of English QA pairs from metadata and translation to Bangla, both processes were conducted using **gemini-2.5-flash**.

3. **Expert Verification** A 1000-sample subset was rigorously validated by two medical specialists, focusing on translation accuracy and positional consistency.

Experiment Setup

Experiment Design: We conduct experiments to evaluate the medical reasoning ability of LVLMs under multilingual settings. Our experiments consist of three main components:

1. **Prompt-Based Reasoning**
 - **Zero-Shot Prompting:** Input = medical image I + system prompt P + Bangla question Q_{BAN} ; Output = open-ended medical answer without task-specific fine-tuning.
 - **Chain-of-Thought (CoT) Prompting:** Explicit reasoning is encouraged by appending “Let’s think step by step” to the system prompt, guiding the model to perform multi-step inference before generating the final answer.
2. **LoRA Fine Tuning**
 - To enhance visual-textual alignment, we employ LoRA finetuning on the frozen model ϕ_M by updating its weights as $W' = W + AB^T$ (where $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$, and $r \ll d$) using inputs I and Q_{BAN} with the loss derived from A_{BAN} .

Experiment Setup

3. Language Variation Analysis

- **Bangla Setting:** Models are queried using native Bangla medical questions to assess multilingual reasoning performance.
- **English Setting:** Corresponding English questions Q_{ENG} are used with the same system prompt P , enabling cross-lingual comparison while preserving semantic consistency.

4. Evaluated Models

- Gemini-2.5-Flash, Claude Sonnet 4, GPT-4.1-Mini, LLaMA-3.2V, Gemma-3, Qwen2.5-VL, LLaVA-1.5, Med-LLaVA, Med-Gemma.

Experimental Results

Models	Chest X-Ray			MedlCaT			Overall (Avg.)		
	Acc	BScore	LAVE	Acc	BScore	LAVE	Acc	BScore	LAVE
Zero Shot Prompt on Bangla QA Pairs									
Closed Source VLMs									
Gemini 2.5 Flash	37.00	82.75	50.80	34.50	81.14	60.08	35.75	81.95	55.44
GPT-4.1 Mini	15.00	72.93	37.70	16.00	73.51	39.50	15.50	73.22	38.60
Claude Sonnet 4	34.50	80.80	43.08	29.00	80.22	47.47	31.75	80.51	45.28
Open Source VLMs									
Llama-3.2V 11B	8.50	73.43	17.50	11.00	81.01	33.50	9.75	77.22	25.50
Gemma-3 12B	29.50	72.62	39.70	39.50	76.70	53.37	34.50	74.66	46.54
Qwen2.5-VL 7B	21.00	75.36	32.85	28.50	75.53	38.30	24.75	75.45	35.58
LLaVA-1.5 7B	8.50	74.43	15.20	14.50	75.14	18.75	11.50	74.79	16.98
Open Source Medical VLMs									
Med-LLaVa 7B	7.00	41.87	12.70	6.00	26.21	30.72	6.50	34.04	21.71
Med-Gemma 4B	4.00	68.91	11.90	18.50	76.02	27.80	11.25	72.47	19.85
CoT Prompt on Bangla QA Pairs									
Closed Source VLMs									
Gemini 2.5 Flash	31.00	83.10	49.33	34.00	82.13	60.58	32.50	82.62	54.96
GPT-4.1 Mini	11.50	69.70	36.15	16.00	72.91	35.70	13.75	71.31	35.93
Claude Sonnet 4	33.50	81.32	43.63	31.00	81.95	51.60	32.25	81.64	47.62
Open Source VLMs									
Llama-3.2V 11B	13.00	73.68	30.17	21.50	82.94	47.18	17.25	78.31	38.68
Gemma-3 12B	31.00	74.26	41.00	40.50	78.72	56.33	35.75	76.49	48.67
Qwen2.5-VL 7B	20.00	74.52	31.85	29.00	75.51	37.15	24.50	75.02	34.50
LLaVA-1.5-7B	1.30	70.44	6.05	2.00	70.58	4.55	1.65	70.51	5.30
Open Source Medical VLMs									
Med-LLaVa 7B	2.60	15.93	14.35	3.27	31.79	32.31	2.94	23.86	23.33
Med-Gemma 4B	12.00	69.21	22.50	28.00	75.18	34.63	20.00	72.20	28.57
LoRA Fine-Tuning on Bangla QA Pairs									
Open Source VLMs									
Llama-3.2V 11B	47.00	92.77	70.70	48.00	89.00	62.78	47.50	90.89	66.74
Gemma-3 12B	44.00	92.13	67.45	48.50	88.57	62.60	46.25	90.35	65.03
Qwen2.5-VL 7B	47.00	92.64	69.75	51.50	89.08	64.60	49.25	90.86	67.18
LLaVA-1.5-7B	42.50	91.55	64.65	46.60	87.43	58.22	44.55	89.49	61.44
Open Source Medical VLMs									
Med-LLaVa 7B	43.00	91.68	65.38	47.50	87.23	58.57	45.25	89.46	61.98
Med-Gemma 4B	52.00	94.06	76.55	50.00	89.56	63.35	51.00	91.81	69.95

Table 1. Model Benchmarking for Bangla with average scores across ChestX-Ray and MedlCaT datasets. Blue highlights the highest-performing model in the Benchmark, while Cyan marks the best-performing models for each metric across the model types.

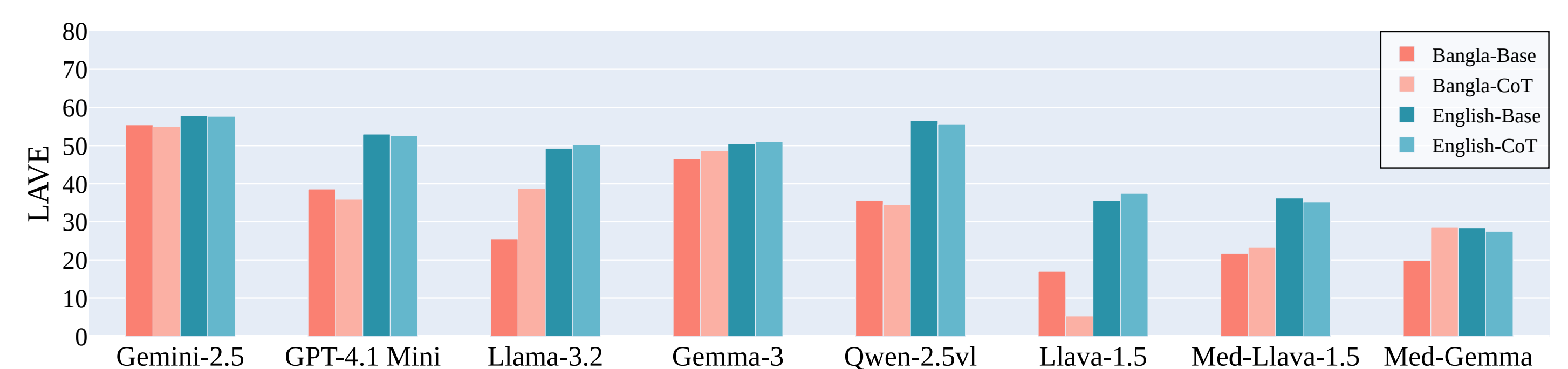


Figure 2. Model comparison of the entire dataset for different models under four different settings: vanilla for Bangla, CoT for Bangla, vanilla for English, CoT for English.

Findings

- **Performance & Language Gap:** Closed-source models (e.g., Gemini 2.5 Flash) outperform open-source models, though all experience severe performance degradation in Bangla compared to English.
- **Diagnostic Failure:** While models handle generalized questions well, they struggle significantly with fine-grained diagnostic reasoning (e.g., specific Conditions or Positions).
- **Prompting & Fine-Tuning:** Chain-of-Thought (CoT) prompting primarily helps general open-source models. LoRA fine-tuning substantially boosts overall accuracy (propelling Qwen2.5-VL and Med-Gemma 4B-it to the top), yet specialized diagnostic limitations persist.
- **Medical Model Limitations:** Despite being tailored for healthcare, domain-specific models like Med-LLaVA exhibited limited cross-lingual capabilities, frequently underperforming general-purpose open-source models in Bangla prior to fine-tuning.